

COMPUTER VISION • PROJECT DEFENSE

Road Detection Using CNN and U-Net

Pixel-level semantic segmentation
for identifying road regions in RGB
road-scene images.

PROBLEM FORMULATION

The project turns road images into pixel-level masks

The model classifies every pixel as road or background, transforming a 3-channel RGB road image into a 1-channel binary road mask.

INPUT → OUTPUT

INPUT

**3-
channel**

RGB road
image

OUTPUT

**1-
channel**

Binary road
mask

Use cases: intelligent transportation, ADAS, autonomous driving, mapping, and scene understanding.

The pipeline covers the full segmentation workflow

01

Download and inspect

Download and inspect paired images and masks.

02

Preprocess and resize

Preprocess and resize the data.

03

Split the data

Split into train, validation, and test sets.

04

Train U-Net

Train a U-Net-style CNN.

05

Evaluate

Evaluate with accuracy, Dice, and IoU.

06

Visualize and save

Visualize masks and save the model for inference.

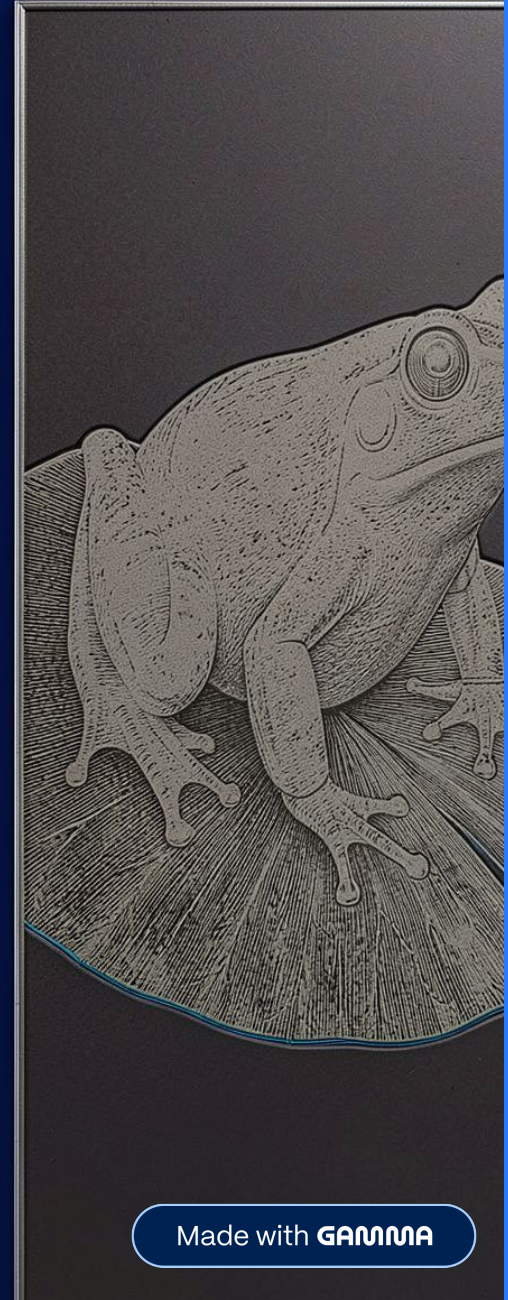
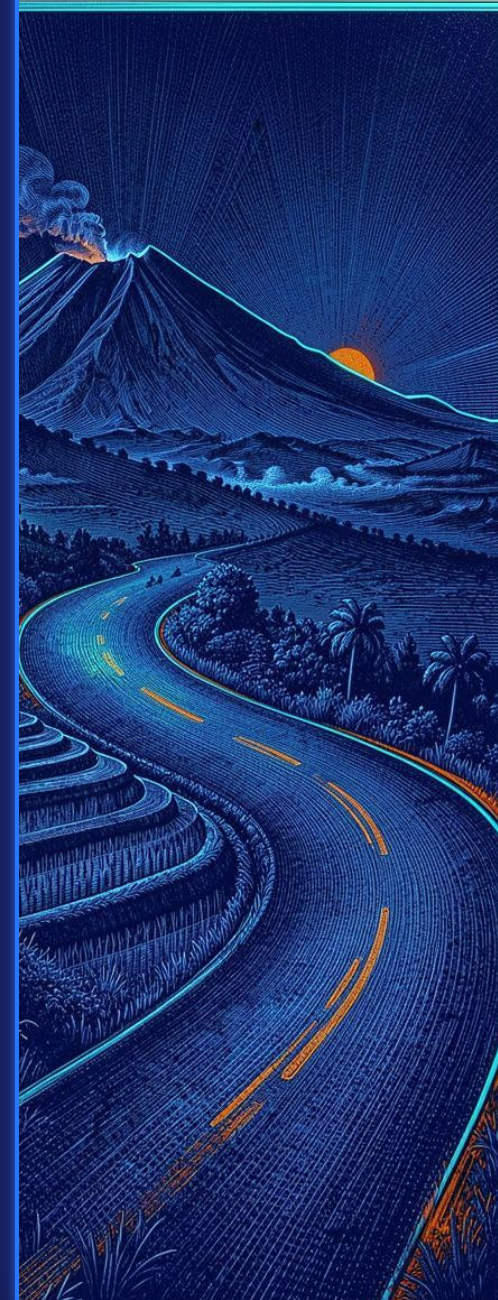
DATASET

The dataset provides
5,442 matched road
image-mask pairs

Dataset: mariahadjmessaoud/road-
detection-dataset

Original image size: $512 \times 512 \times 3$
RGB; ground-truth masks: $512 \times 512 \times 1$
grayscale.

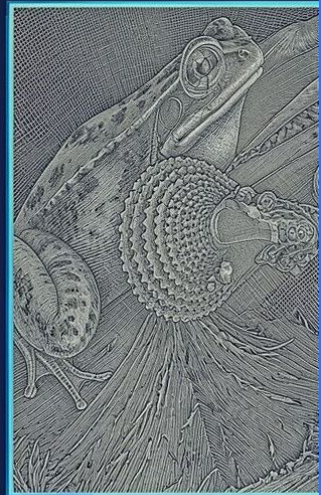
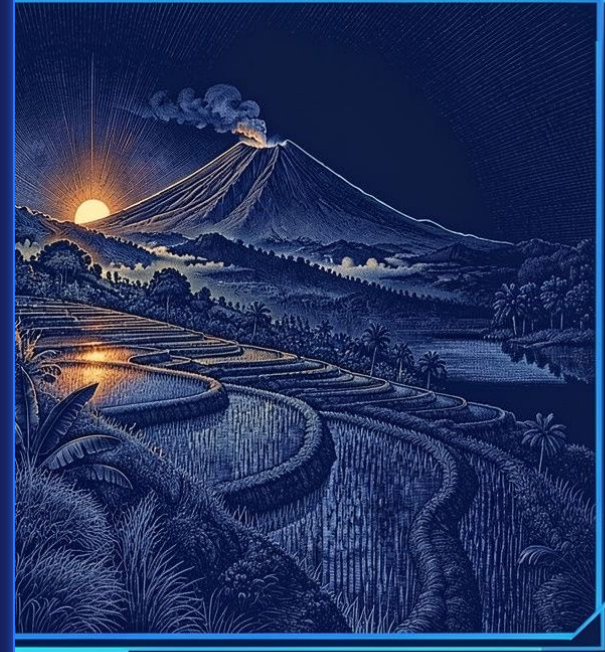
Images and masks use matching
filenames such as road_0001.png;
training data was selected from the
train/images and train/masks



PREPROCESSING

Preprocessing makes the data consistent and memory-efficient

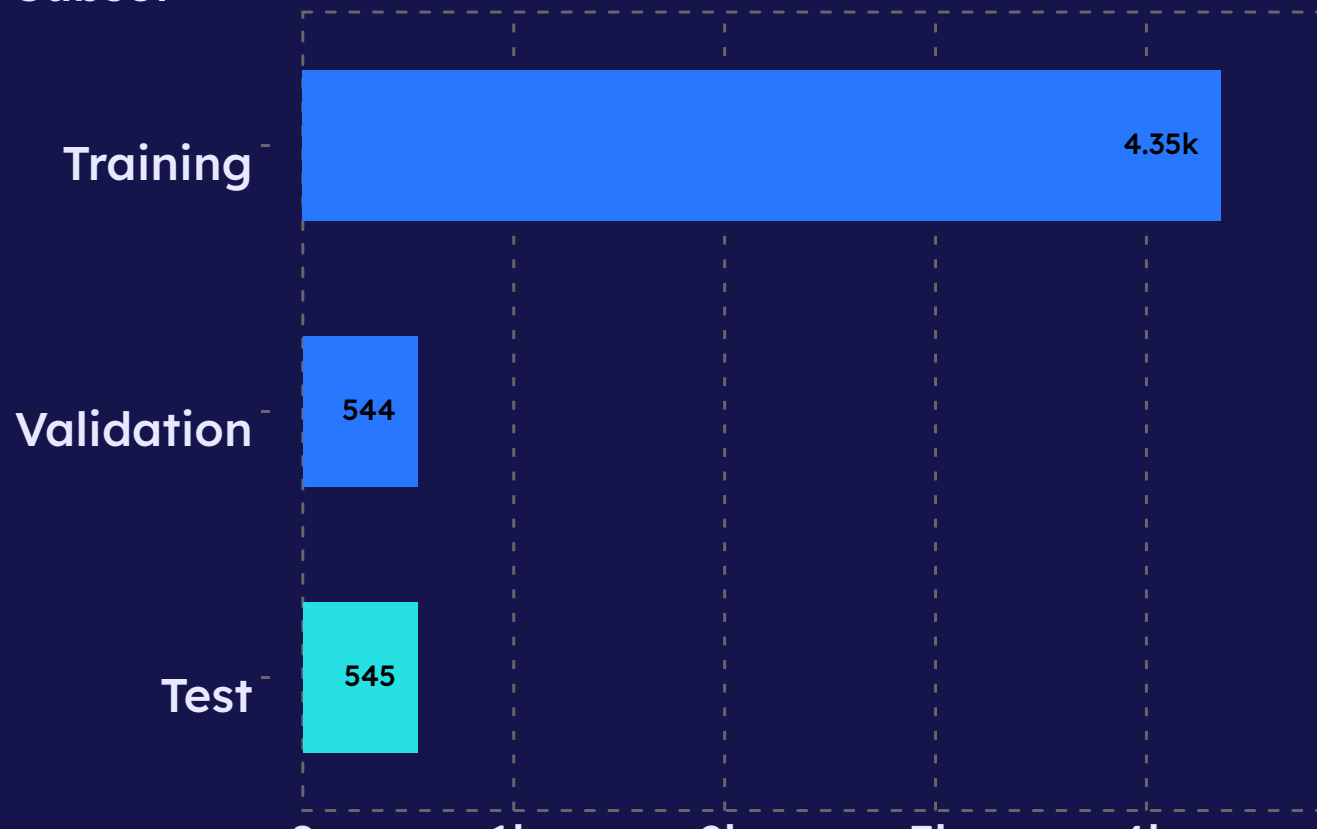
1. Load RGB images and grayscale masks
2. Resize both from 512×512 to 256×256
3. Normalize image values to approximately 0–1
4. Normalize masks and threshold them at 0.5
5. Serve batches of 16 through a TensorFlow Sequence generator



The split protects evaluation from training bias

Most paired samples support learning while validation and test sets provide independent checks.

Subset



Training: 4,353 samples — 80%

Validation: 544 samples — 10%

Test: 545 samples — 10%

Fixed random_state: 42 for reproducibility.

Training data is shuffled

MODEL DESIGN

U-Net combines context with precise road boundaries

The encoder learns richer features as spatial resolution falls. The bottleneck captures scene context. The decoder restores resolution, while skip connections



ARCHITECTURE DETAIL

The architecture grows from 32 to 512 feature channels



Decoder

Upsamples through transpose convolutions.

Convolutional blocks

Two 3×3 convolutions, Batch Normalization, and ReLU.

Output

OBJECTIVE AND METRICS

Combined loss balances pixel accuracy and region overlap

PIXEL ERROR

Binary Cross-Entropy

Penalizes pixel-level classification errors.

Accuracy

Useful, but can be inflated by correctly predicted background pixels.

REGION OVERLAP

Dice loss

Improves overlap between predicted and true road regions.

Dice

Measures overlap between predicted and true road regions.

COMBINED OBJECTIVE

BCE + Dice

Balances pixel accuracy and region overlap.

IoU

Intersection divided by union.

TRAINING RESULT

Training improved overlap quality across 10 epochs

Validation Dice improved from 0.5375 in epoch 1 to 0.6319 in epoch 10. The best recorded validation loss at the end of the run was 0.46437.

TensorFlow 2.21.0 with Keras · Adam · learning rate 1e-4 · batch size 16 · EarlyStopping, ReduceLROnPlateau, and ModelCheckpoint.

